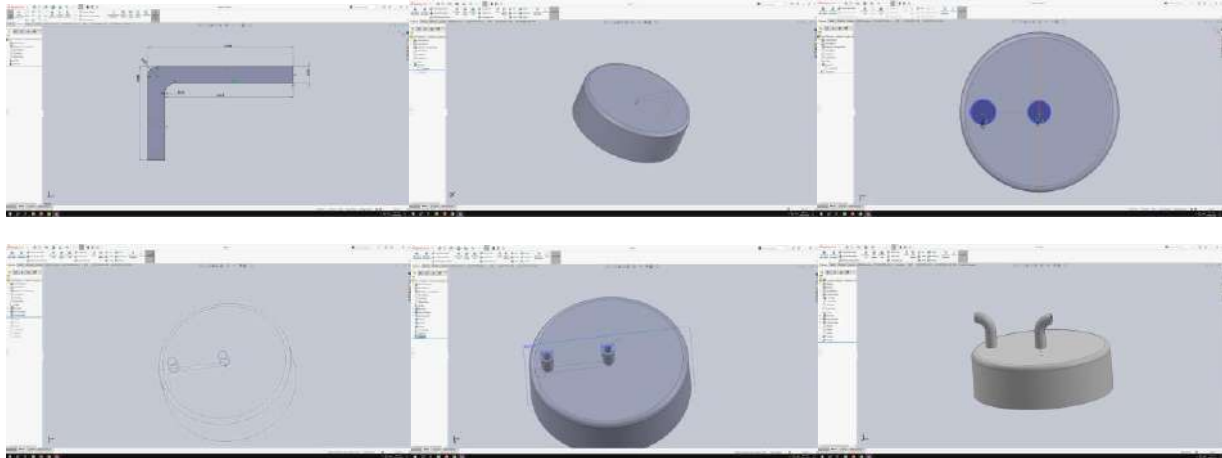


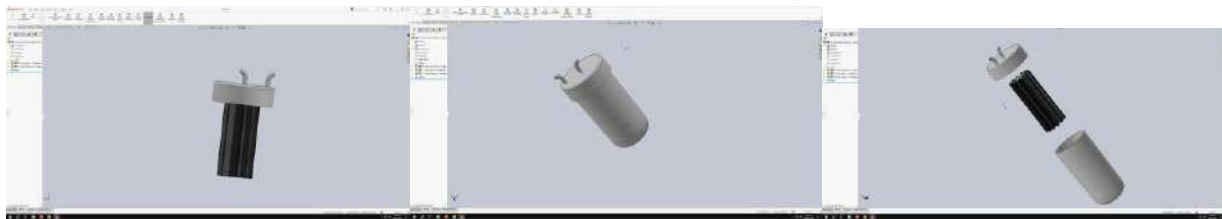
Sub-Part – Pre-Filter Lid and Connecting Pipes | Assembly – Prefilter

Joshua Sutcliffe

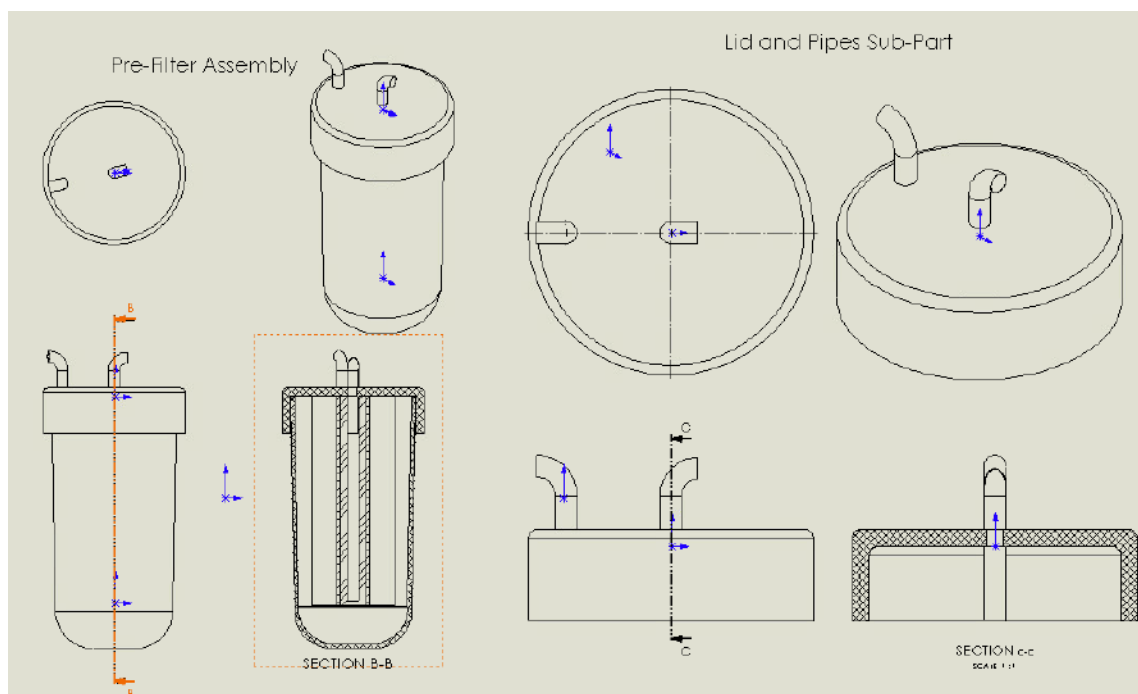
1) Sub-Part



2) Assembly



3) Drawing views

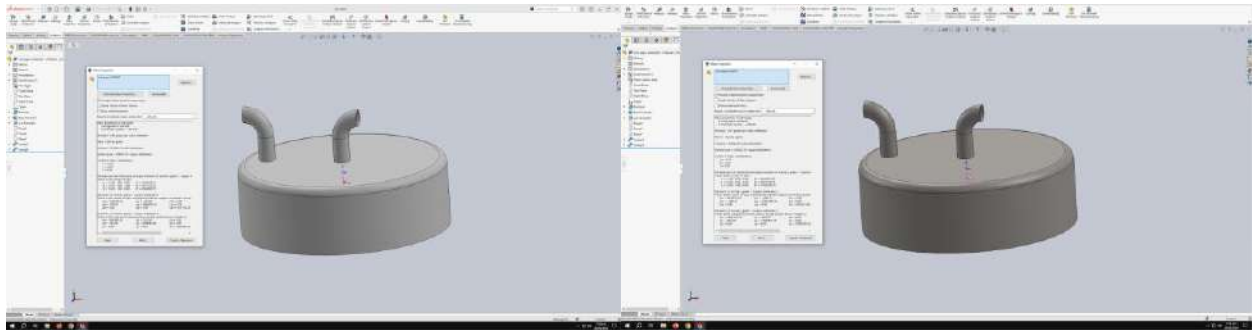


4) Mass Properties

Sub-Part:

Volume: 452,942.54 cubic millimeters

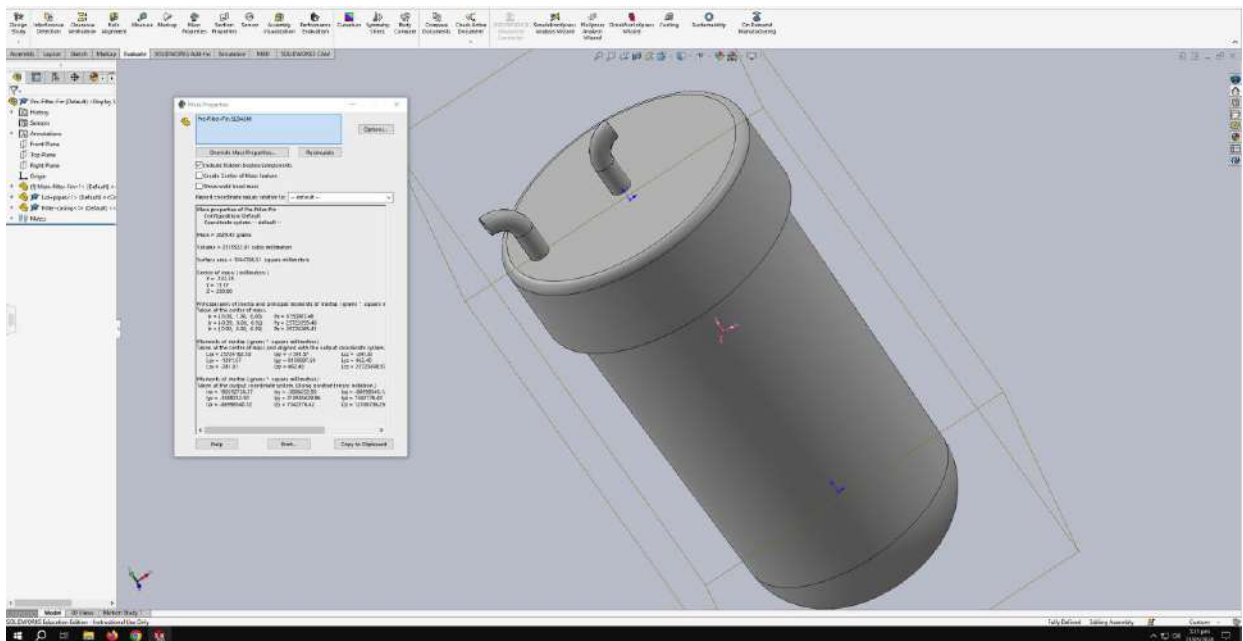
Mass: Steel 3532.95 g, compared to PVC 588.83 g



Assembly:

Volume: 2,135,522.91 cubic millimeters

Mass: 2629.43 grams



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Week 9

Does our solution meet the criteria for assessment 2?

Technical understanding:

- The design criteria was developed to suit the needs of the Saibai community. They have their own unique problems in the water and sanitation area, which only certain solutions will be able to be implemented due to the local limitations like resources, environment and community need/want.
- In assessment 1 we compared each of our 5 designs to each other and how they stacked up to the design criteria. Therefore we believe we selected the best design for the criteria.
- Our final design fits the technical, community and sustainability understanding required of this project. Our goal was to create a new way of producing potable water that wasn't reliant on the regions rainfall. This meant that there were already some limitations based on the surrounding geographical location. Analysing this we decide on a desalination as the method for creating potable water. From there we need a solution that provided for the current community size but also was scalable, used as many local resources and could be maintained by someone from Saibai island, and could be implemented without using more power than needed. From these and other considerations we came to the solution of a small scale Nanofiltration systems. This design can provide enough water for the current community and has the ability to be scaled by increasing the number of filters or pump flow rate, can be implemented using a limited number of offshore components that can be transported easily, and in combination of using brackish water, will use a small amount of energy. Nanofiltration is technically sound due to a few factors. First, the selective filtration, the filtration membranes have pores on a nanometer scale allowing them to selectively filter elements like salts and bacteria while allowing water to pass through. Secondly, there is a low chemical dependency, this means compared to other design like reverse osmosis, nanofiltration has reduced environmental and operational costs. Lastly the modular robust design, this means the system is able to withstand fluctuating water feed (both quality and quantity) and operating conditions making it scalable for different water demands ensuring the consistent performance over time.

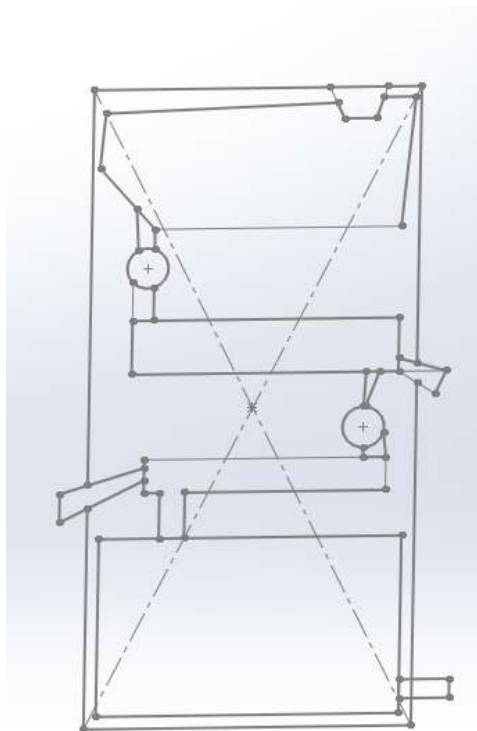
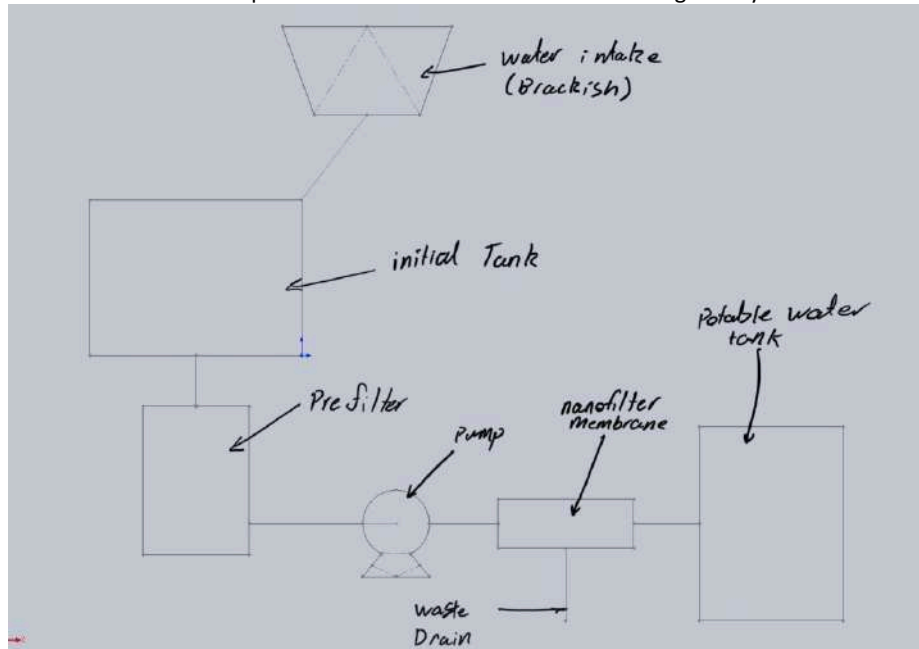
Our design could do with a few improvements and specifications to things like the overall design layout. Currently our design is to conceptual, there are no specification to things like input capacity, maximum output, energy used, etc. Thus to understand which parts of the design need improvement we will need to create more of a practical plan.

Week 10

We had an extra meeting on Wednesday where we came to the conclusion that we need to create more sketches to further our understanding of what we need to design in solid works, we have a current part list at this point of: Nano filter membrane, pump, water inlet, pre-filters, and then general body/pipes.

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I drew this sketch to help me better understand what would be a good layout:



I am currently trying to decide if we will need an actual pre-filter or just something integrated into the water intake that stops solids and other clumps getting into the nano filter. Gopani answers the question "Why Pre-Treatment Of Sea Water Before Desalination Is Important" by stating that it helps remove impurities in the sea water prolonging the life of the filter. They also argue the case that as seawater becomes more and more polluted from human activities the more important a pre filter will become as the build-up of pollutants such as carbon will drastically increase. From this information I believe we should have a separate prefilter, it is now about what type of pre-filter it should be and what its target impurity will be. The water

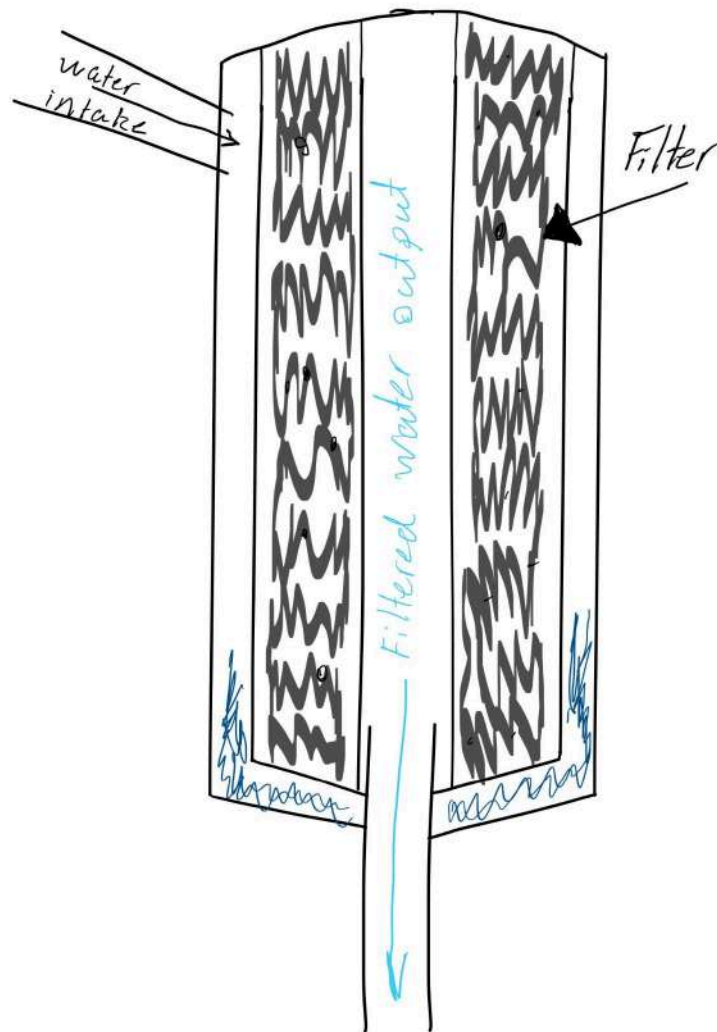
intake will still need to have some kind of filtration as it should be the primary way sand, mud and other large solids don't enter the system.

From research I have done it is best that we remove large particle up to $5\mu\text{m}$ size (GmbH, n.d.). This will remove a large amount of the particles in the brackish water. These could be removed by the nano filter but to prolong the life of the nanomembrane using a prefilter of this size will get rid of most of the particles that could damage it.

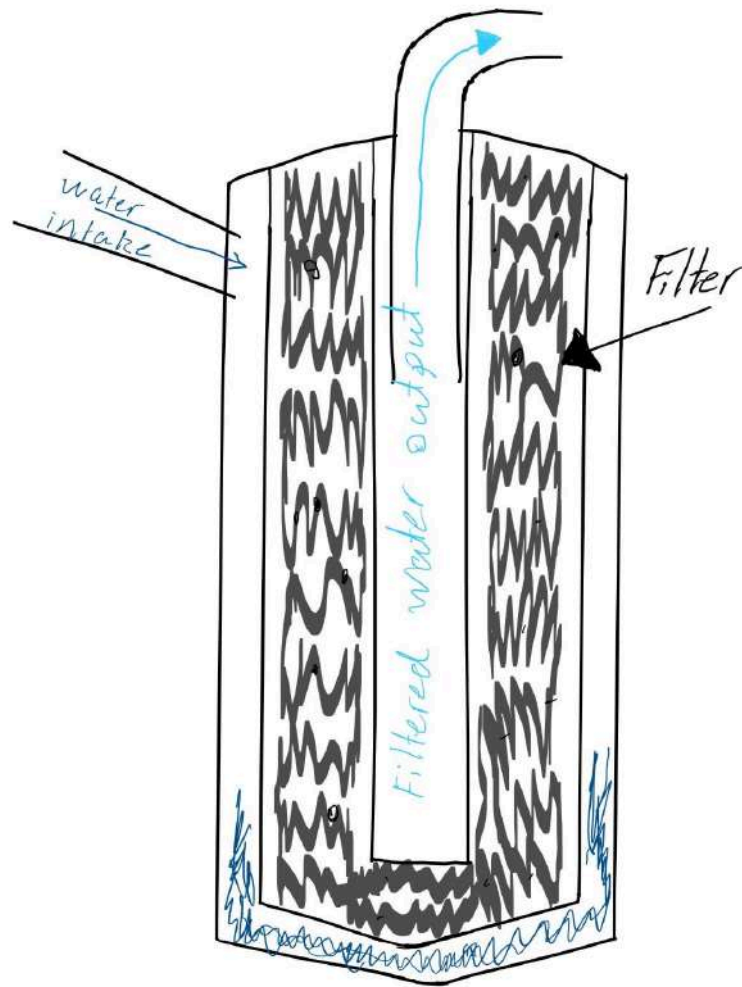
How do Pre-filters actually work? As water passes through the filter the larger particles will get stuck on the membrane and not be able to pass through while the water will easily be able to pass through, continuing to the next part of the system.

For my design of the pre-filter, I will have the intake water go in to an area surrounding the filter, and for it to escape the system it will have to go through filter to the centre of the system where it will be taken out of the filter.

1) Extracted from the bottom



2) Extracted from top



Of the 2 designs here I will choose the one that fits our design and layout best. From here I will develop the part further, creating a sketch of the filter itself and the prefilter housing.

- Systems, G. P. (2024, May 11). Desalination - Advanced pre-filtration systems to solve and challenges. Gopani Product Systems Private Limited.
<https://www.gopani.com/desalination/#::~:~:text=Pre%2Dfiltering%20protects%20the%20RO,increasing%20at%20an%20alarming%20rate.>
- GmbH, C. C. (n.d.). Filtration for desalination. 2024, © Boll & Kirch Filterbau GmbH.
<https://www.bollfilter.com/applications/treatment-of-water-systems/filtration-for-desalination>
- Admin. (2023b, June 1). Understanding the role of Pre-Filters in carbon filters. Blog.
<https://floraflex.com/default/blog/post/understanding-the-role-of-pre-filters-in-carbon-filters>

<https://www.svb24.com/en/katadyn-pre-filter-for-desalination-pumps.html>

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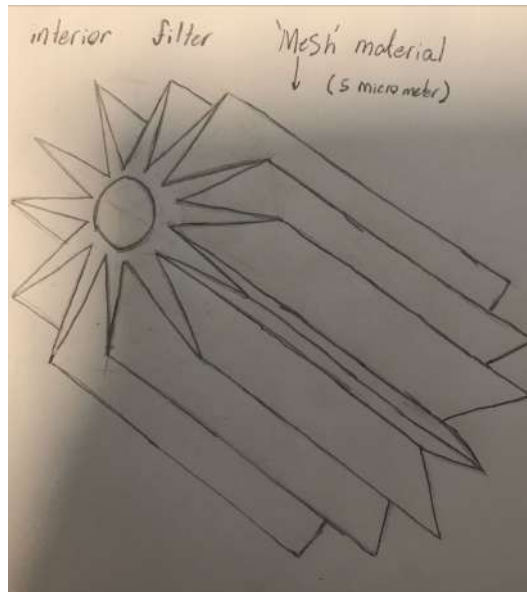
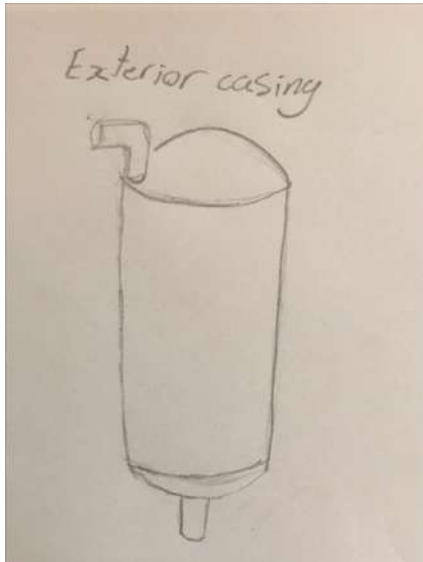
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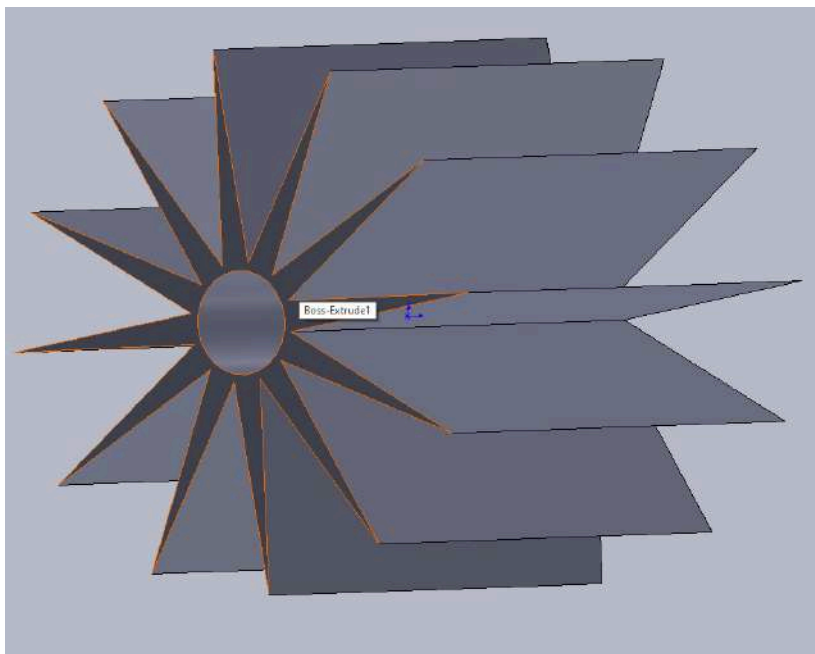
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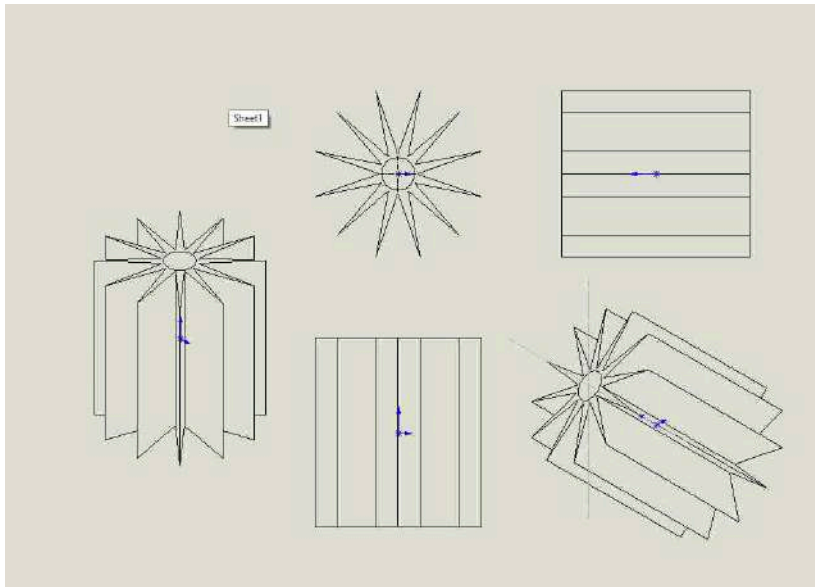
Week 11

After last week's research I will be continuing with my part design, starting with a few sketches.



After creating these sketches I quickly mocked up the Interior filter design I SolidWorks:





After meeting with my group it was decided that a vertical casing would be the best fit.

After doing some research into the materials I should use I decided to have a dual purpose filter. This filter will use both activated carbon (Wfa & Wfa, 2021) and microfilter technology to make the water more drinkable.

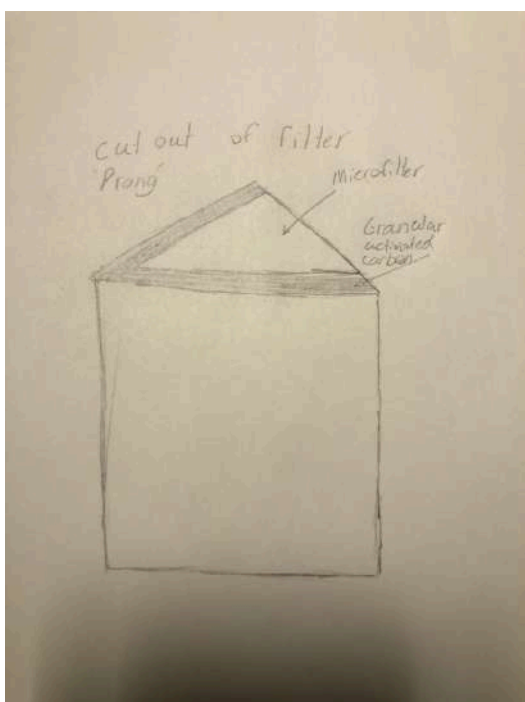
Micro filter (GearSkeptic, 14/2/2022)

- Use a small micron rating (5 micrometers) to filter out bacteria and some viruses—at this point there should be no huge molecules like sand and mud.
- This is need to make sure there are no microorganisms (bacteria, viruses, etc.) living in the water

Activated carbon filter (GearSkeptic, 23/5/2022)

- Uses granular activated carbon (activated charcoal) which has porous structure that traps contaminants. GAC often has a vast surface area often exceeding 1,000 square meters per gram.
- Is need to remove things like heavy metals (lead, mercury, etc.), Volatile Organic Compounds (benzene, trichloroethylene (TCE), etc.) and Microplastics.

To Integrate this design into one filter, there will be a layer of activated carbon surrounding the micro part of the filter. This will mean that the water will first pass the GAC before going through the microfilter.



At this point I have started on my Prefilter development documentation.

Wfa, & Wfa. (2021, July 22). How does charcoal purify water? | Water Filters Australia. Water Filters Australia |. <https://www.wfa.com.au/charcoal-water/>

GearSkeptic. (2022, May 23). Backcountry Water Treatment - Part 5: Activated Carbon Filters for Chemicals and Viruses. YouTube. [Backcountry Water Treatment - Part 5: Activated Carbon Filters for Chemicals and Viruses](#)

GearSkeptic. (2022, February 14). Backcountry Water Treatment, Part 3: Microfiltration. YouTube. https://www.youtube.com/watch?v=IJOiCztnXfY&list=TLPQMjkwNTIwMjSh7mBb1QrQ_w&index=31

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As an individual, I could have not let things stack up till the last minute but I would say this is more a point for the individual work rather than the group project as I kept up to date with the group project. I would say I contributed more than or equal to my expected contribution with the individual project, as I helped with the initial creation of the PowerPoint-specifically researching the challenge set out for our area and the key functions and attributes. I also completed my own individual plan for a Brackish water desalination to a high quality; explaining the key design points, Maintenance/costs, energy effects, efficiency, and total impact on the local community. I then after completing my design helped pick the best solution, Nanofiltration, and begin work on finalizing the project by outlining the remaining categories: a more in-depth system overview and the individual concepts for the work plan. Later on, I then help complete these 2 section to a high standpoint. Overall my individual contribution to assignment one was one of both good quality and quantity.

As a group, I believe our final submission was of great quality. We worked well as a team but there is room for improvement were our processes were to an average standpoint. We could have better organized each individuals personal contribution other than the sketch as after we completed this it was work on it if you can so the final contribution for each member would be skewed. This can be easily rectified by assigning each member their own equal contribution to work on. Our meetings were also okay, most of the time every member was and what needed to happen at these meetings happened. Our final decision for the design area we have chosen fits the challenge well, there are just parts of the design that need to be fleshed out to become more comprehensive.

For the upcoming assignment 2 I am well aware of what is expect of us as an individual and as a group. Each of the 14 steps that need to be added to the final presentation, including the individual CAD design, make sense and will be completed. We have looked over the project and grading criteria and will assign each member a task to complete on top of their CAD drawing.

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Week 12

I have started to complete my design in SolidWorks CAD and have created another document for detailing my decisions and detailed design process of my part of my groups small scale desalination plant.

For my material selection I have gone with activated carbon and polypropylene (PlumbingSupply.com®, n.d.) for the actual filter itself and for the casing and pipes, I have used PVC. The material for the casing and pipe was between PVC and steel; both fit the purpose of housing the filter and transporting water to and from the system.

This week I completed the final design of my pre-filter; completed the CAD process of creating the parts and assembling them.

For our group report I noticed that the structure and layout was all over the place and didn't have all the sections needed. So I went in and changed the structure and added the sections we needed. This involves creating the contents page, space for the executive summary, list of figures, introduction, lifecycle issues, conclusion, recommendation for further work, references, appendix, and personal reflections. I also added the contribution table. I then completed the design method section by writing why there was a need for our system and structuring out the elements we completed in our first presentation assignment.

I then completed my all my sections within the document (list of figures, design method conceptual, lifecycle issues, appendix 1, and personal reflection).

I also added a statement to the conclusion.

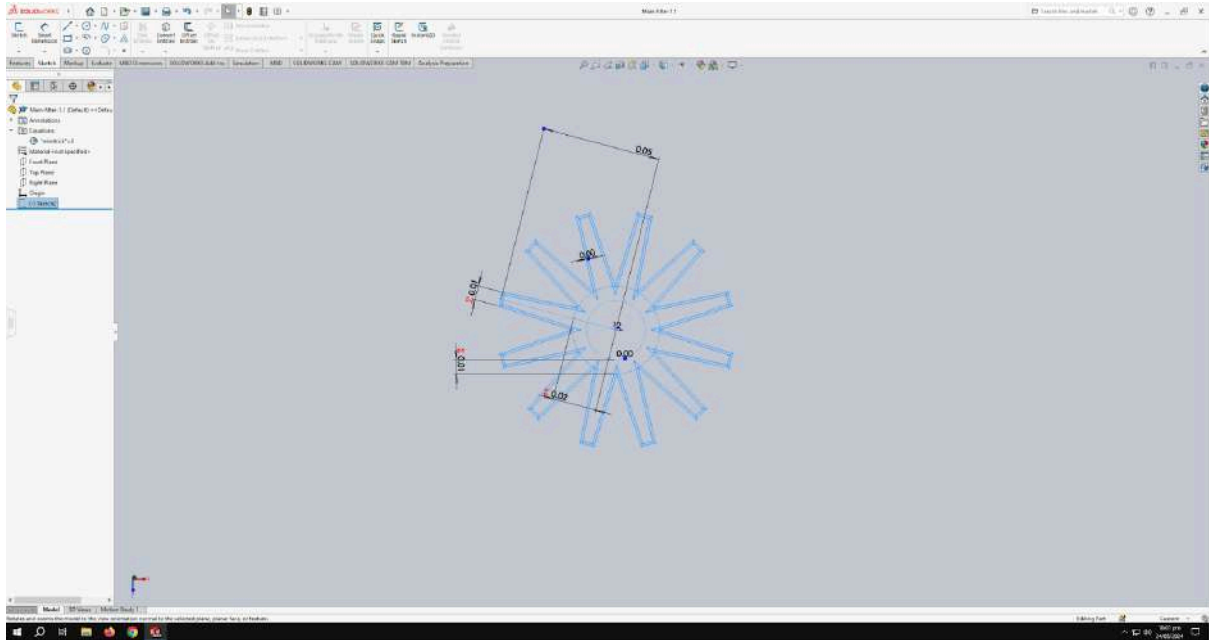
PlumbingSupply.com®. (n.d.). How to choose a water filter. PlumbingSupply.com® - the Premier Online Plumbing Supplier Since 1995. <https://www.plumbingsupply.com/water-filter-buying-guide.html>

Development of my part of the Desalination system

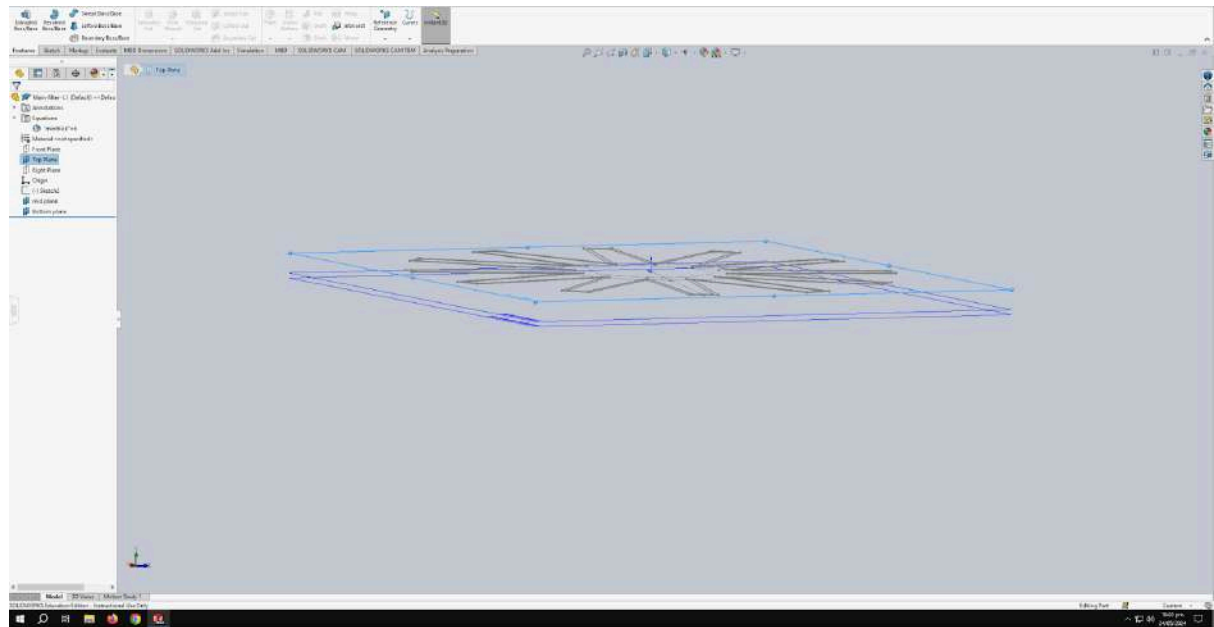
I have made an assembly rather than a part, as it has made the design process more logical and straightforward

Part 1 - Main Filter:

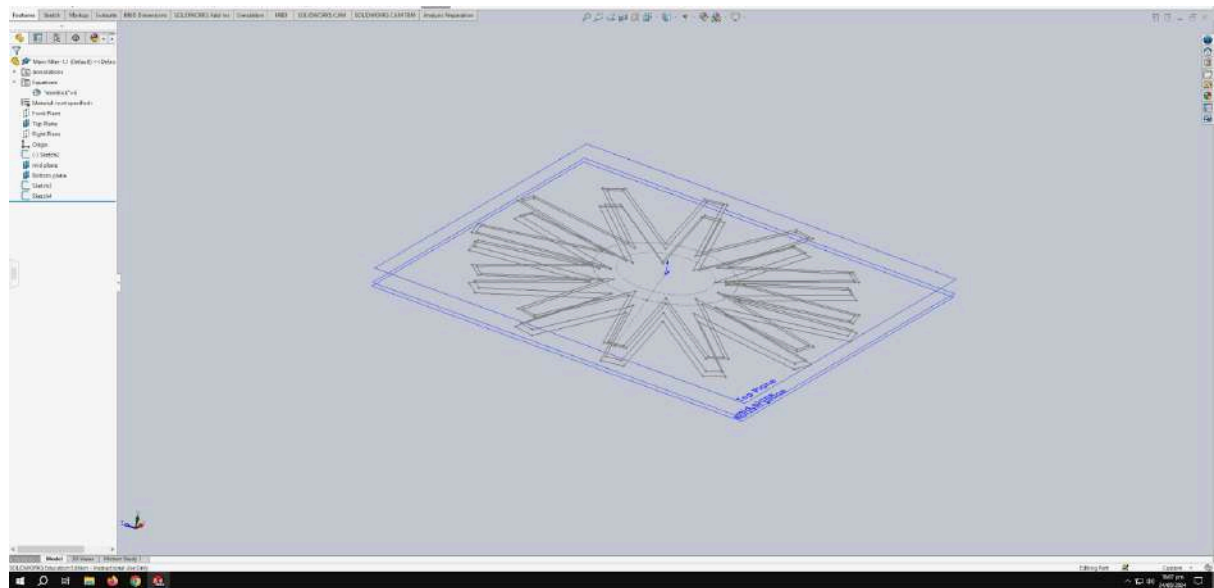
- 1) Created the main sketch. I used this shape as it has a larger surface area available for the water to filter through than just a regular circle, square, or hexagon shape. It has a diameter



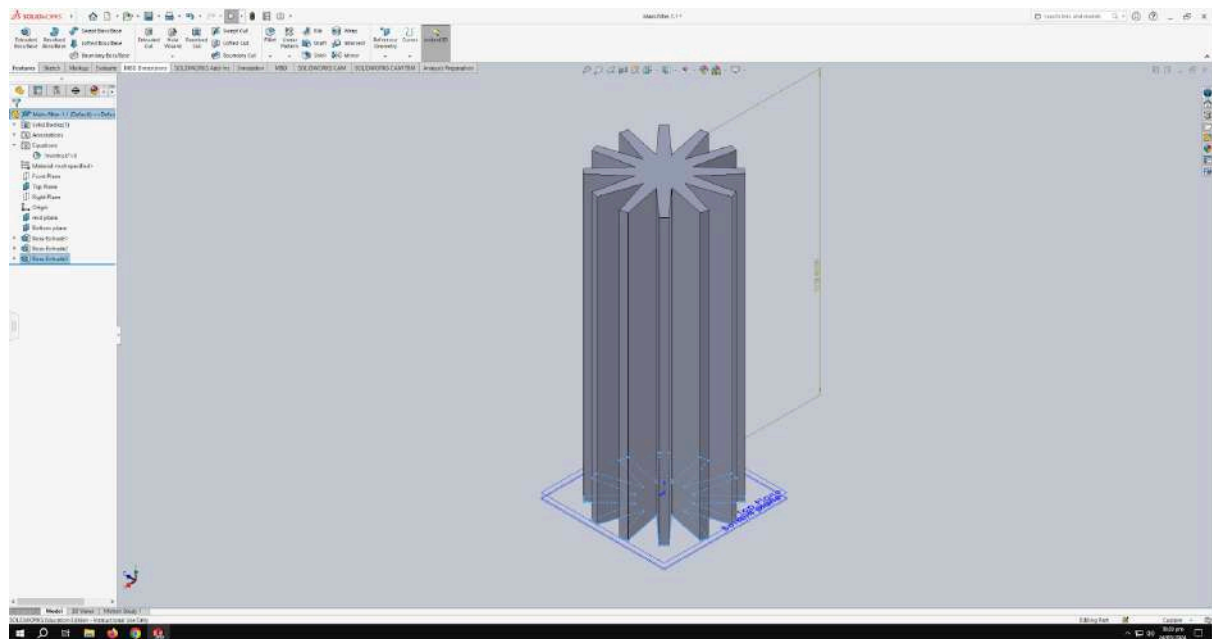
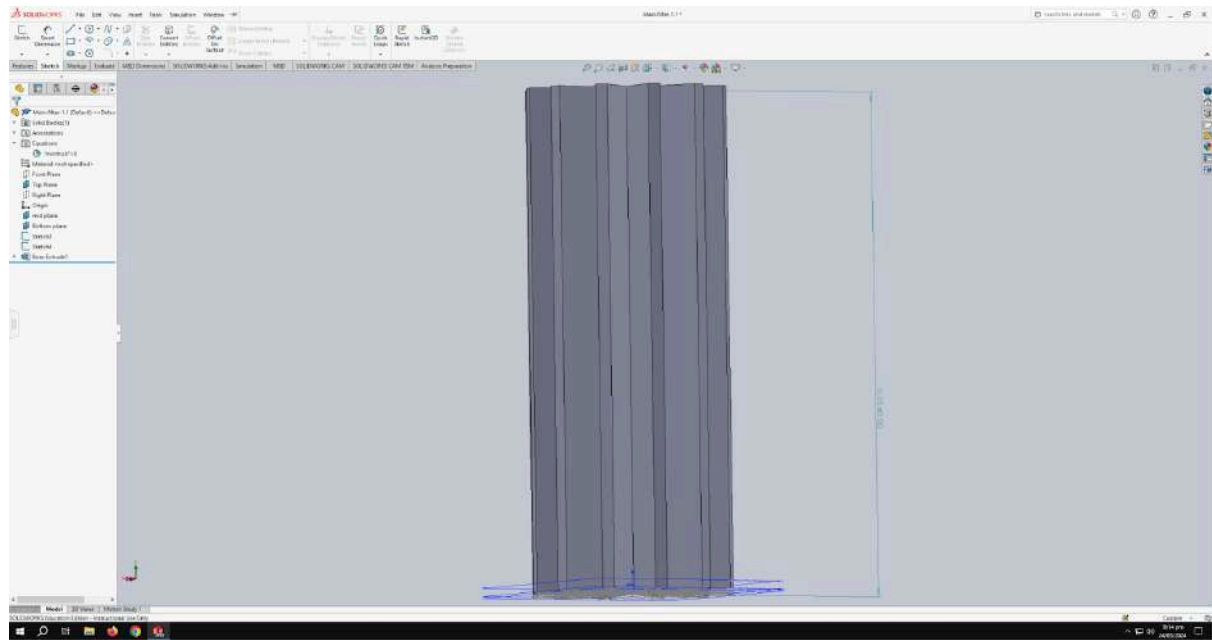
2) I Created 2 offset planes for another 2 sketches



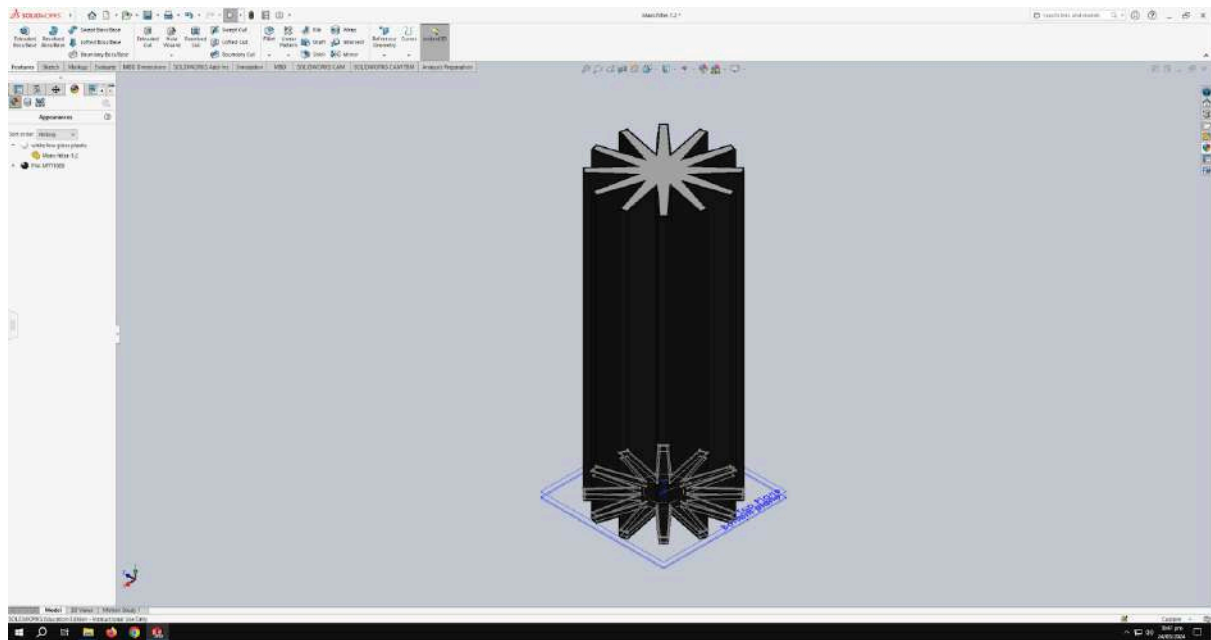
3) Convert entities to new planes



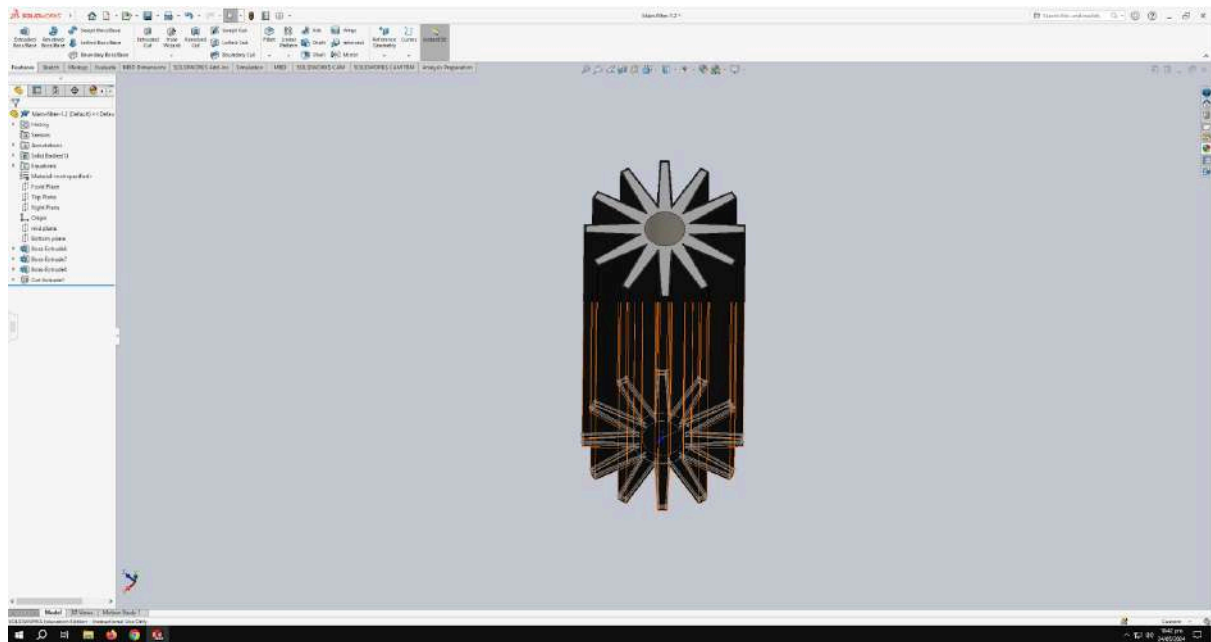
4) Extruded all sketches out



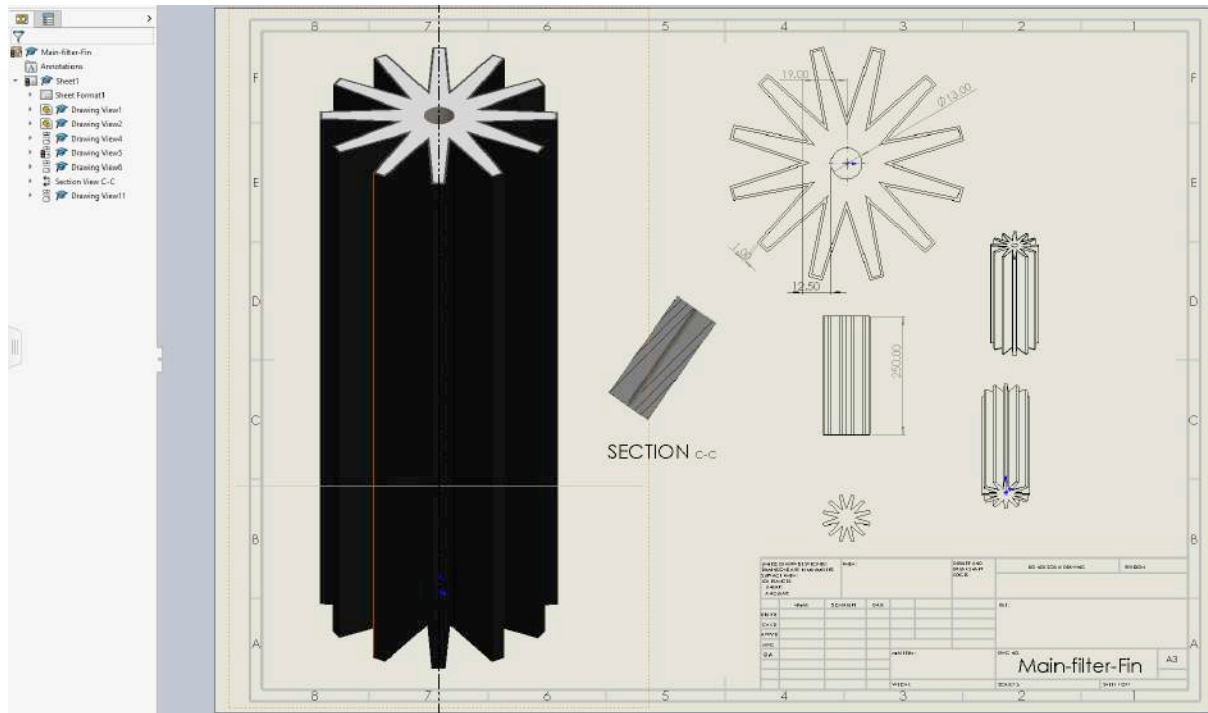
5) Changed the appearance of the bodies to distinguish between carbon filter elements and microfilter elements



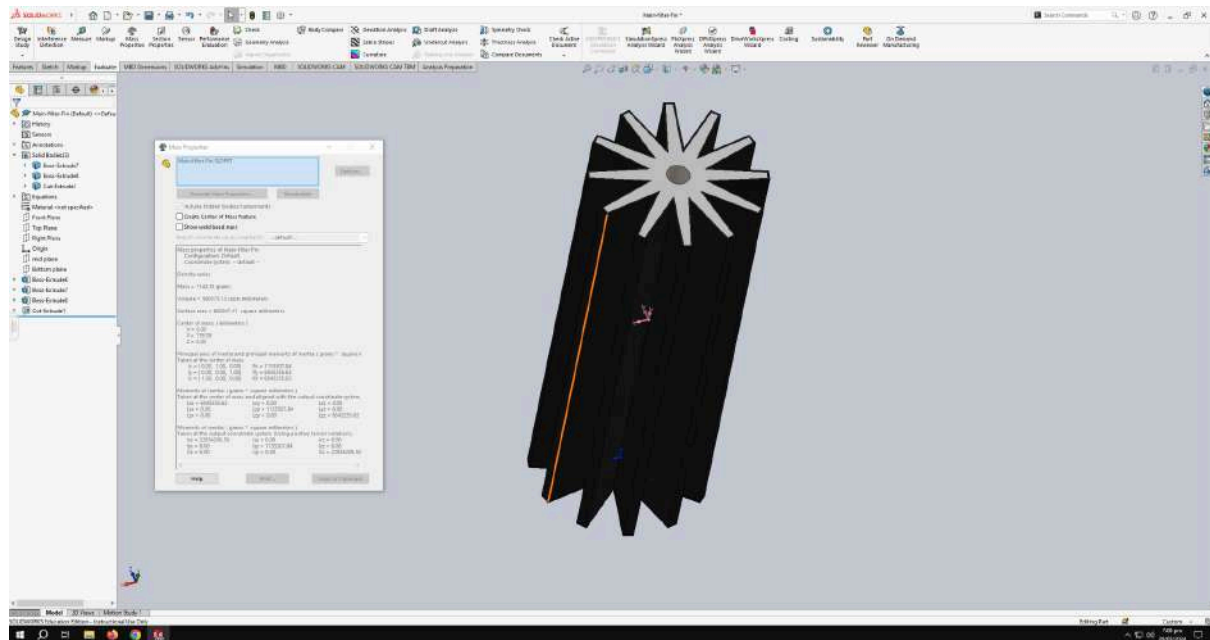
6) Added an extrude cut for where the pipe connects.



7) After refining after a group discussion, the final measurements are a diameter of 100 mm, a minimum thickness of 6 mm for the filter, and a total length of 250 mm. The pipe's outside diameter should also be 13 mm.

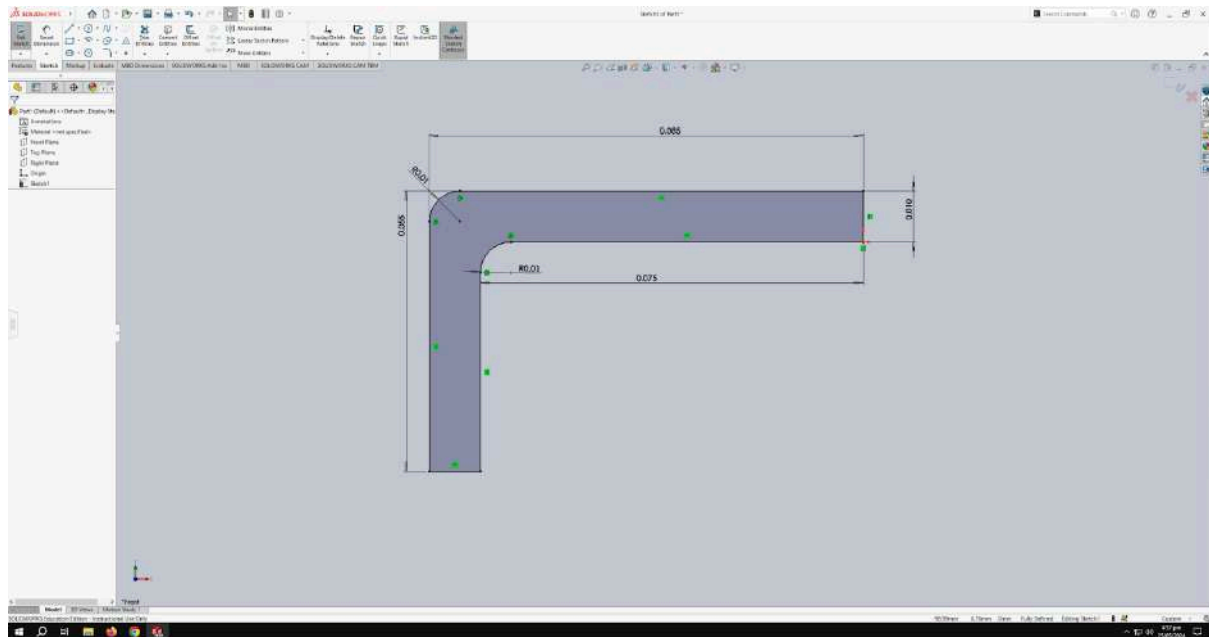


8) Changing the materials to be more accurate, the closest material I could find in the solid works library to activated carbon was VCB-20 Carbon Cloth, so that evaluations will be based on this material. The same is said for polyporpylene, as the material is not available in solidworks, I used polyethylene cross-linked instead.

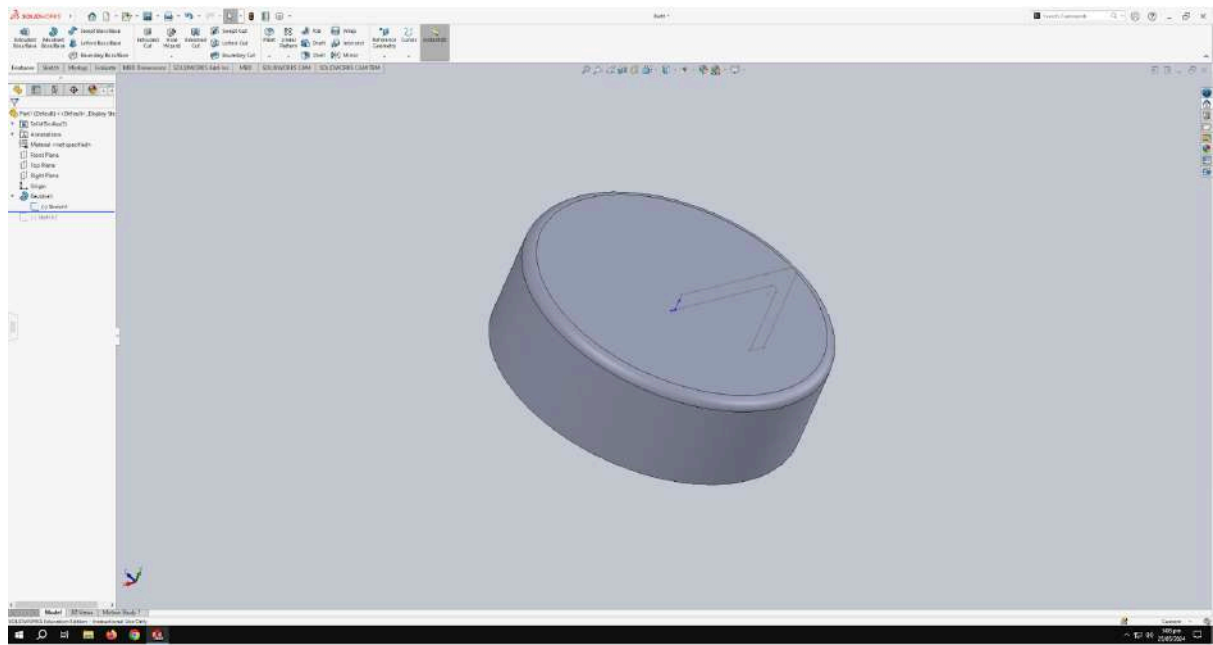


Part 2 - Lid and Pipes

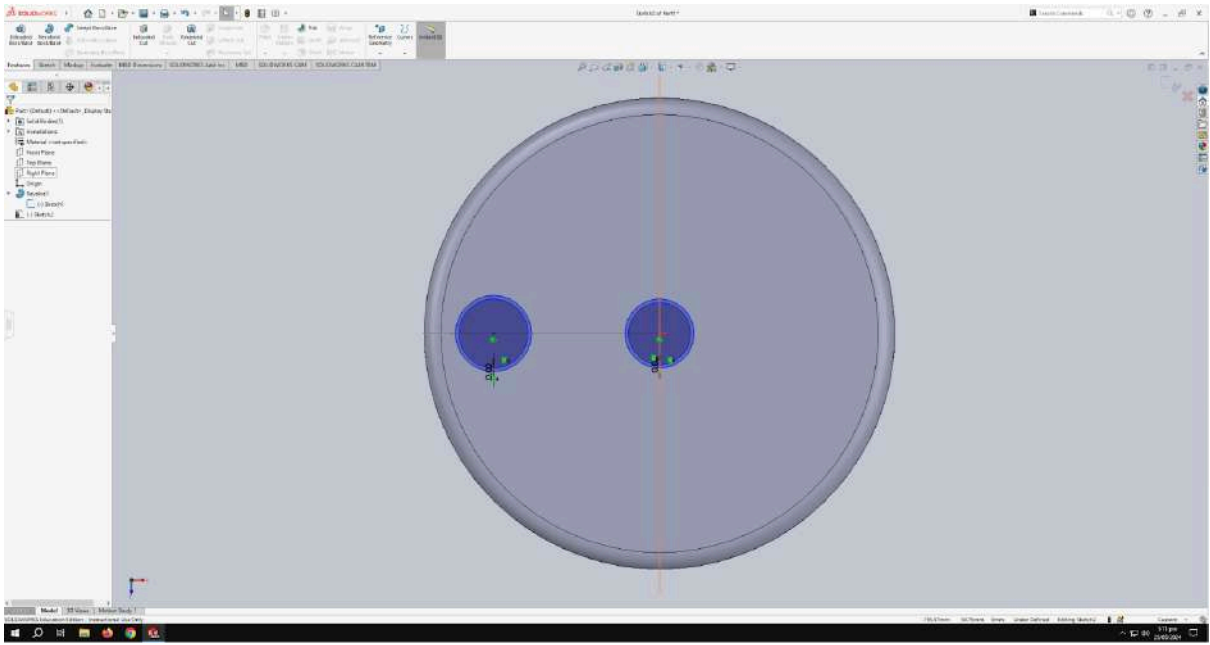
1) Sketch half the lid shape in 2d - used a Fillet



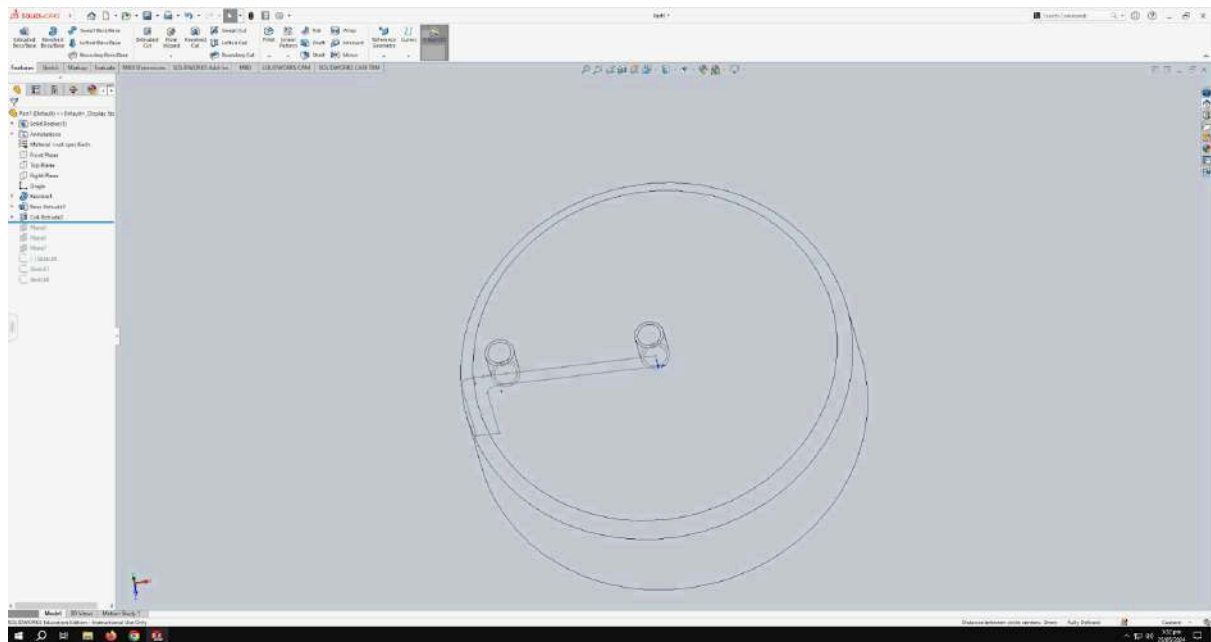
2) Revolve boss/base to create a 3D lid shape



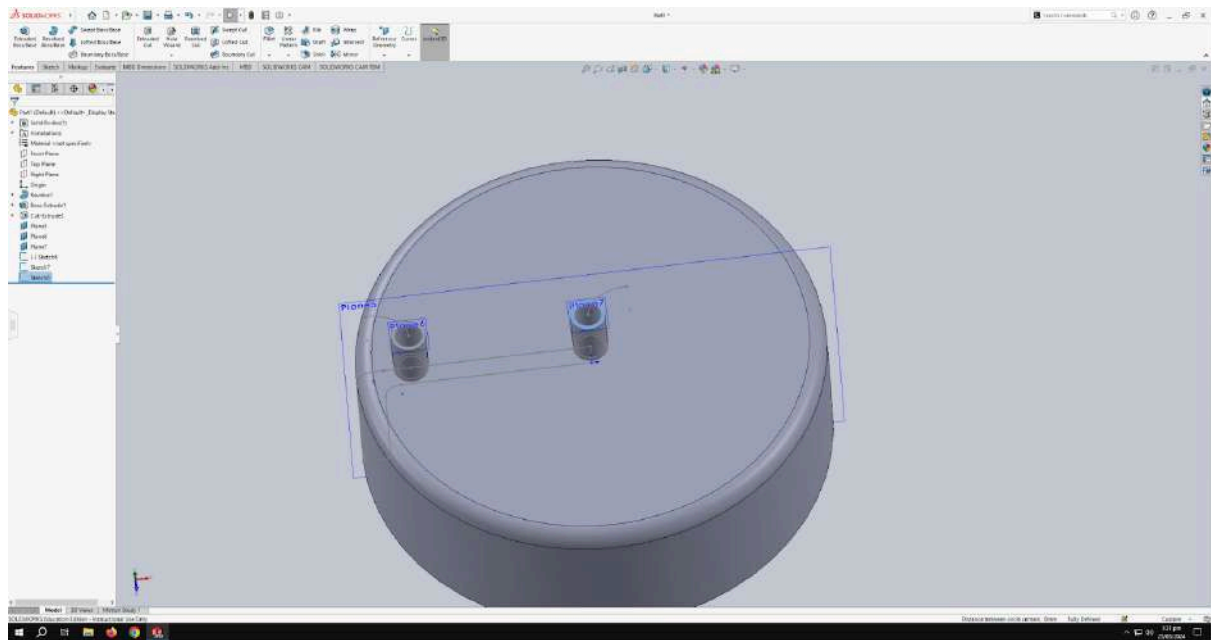
3) Create a sketch for where the pipes will go



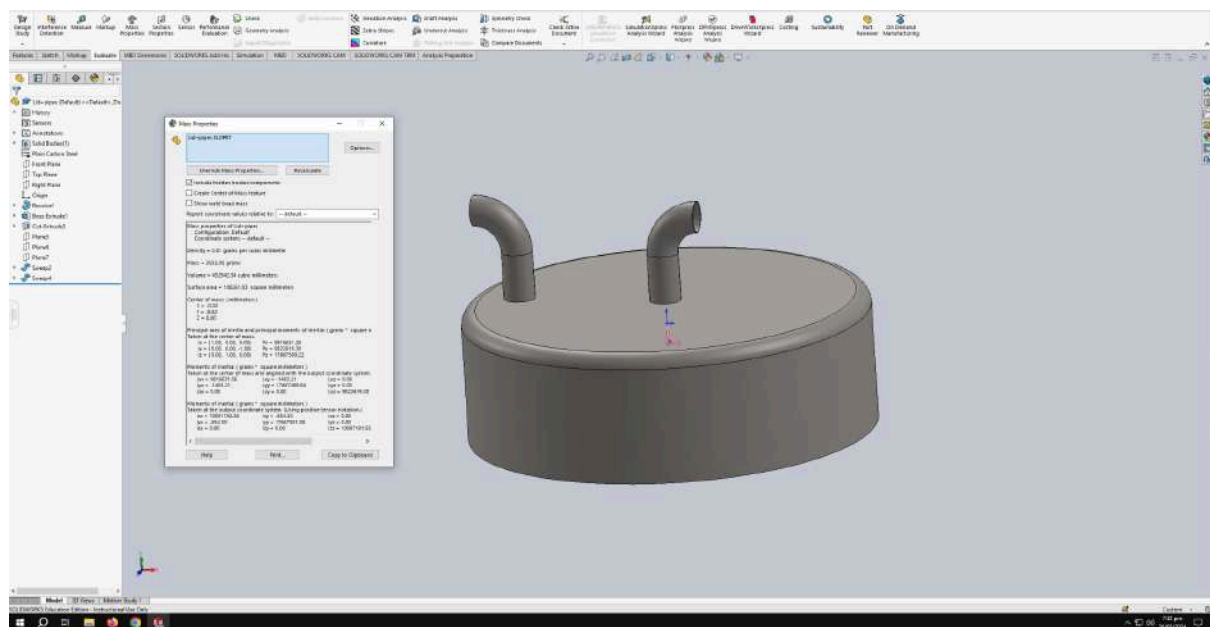
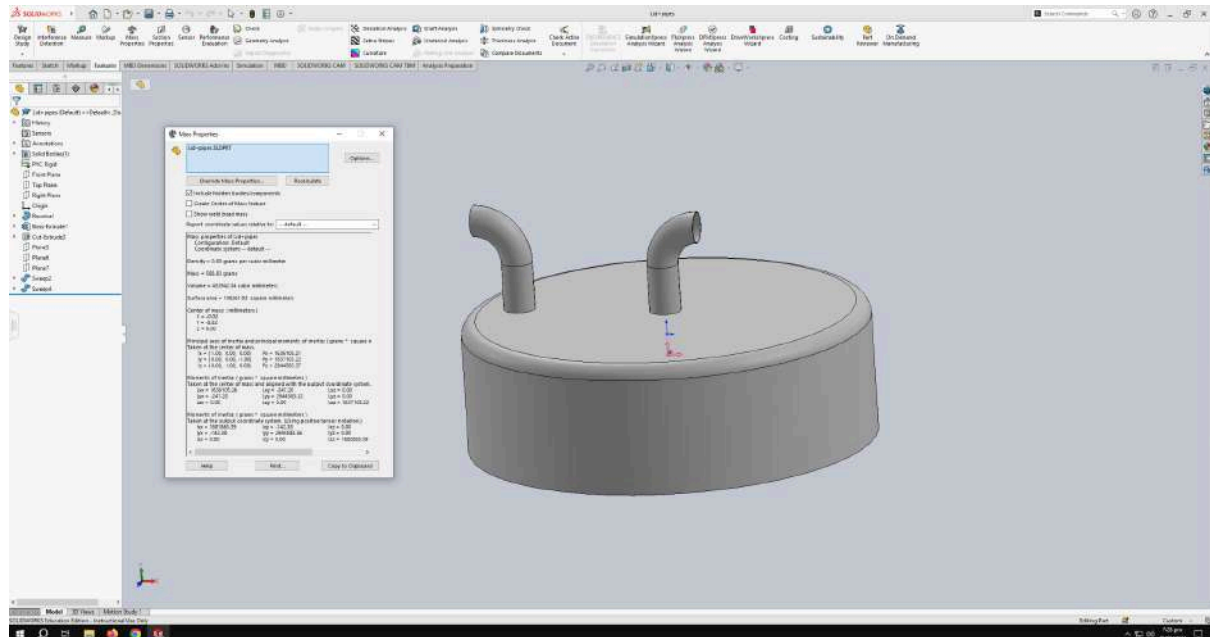
- 4) Extrude out the pipe shape, extrude cut to make them hollow and pass through the lid



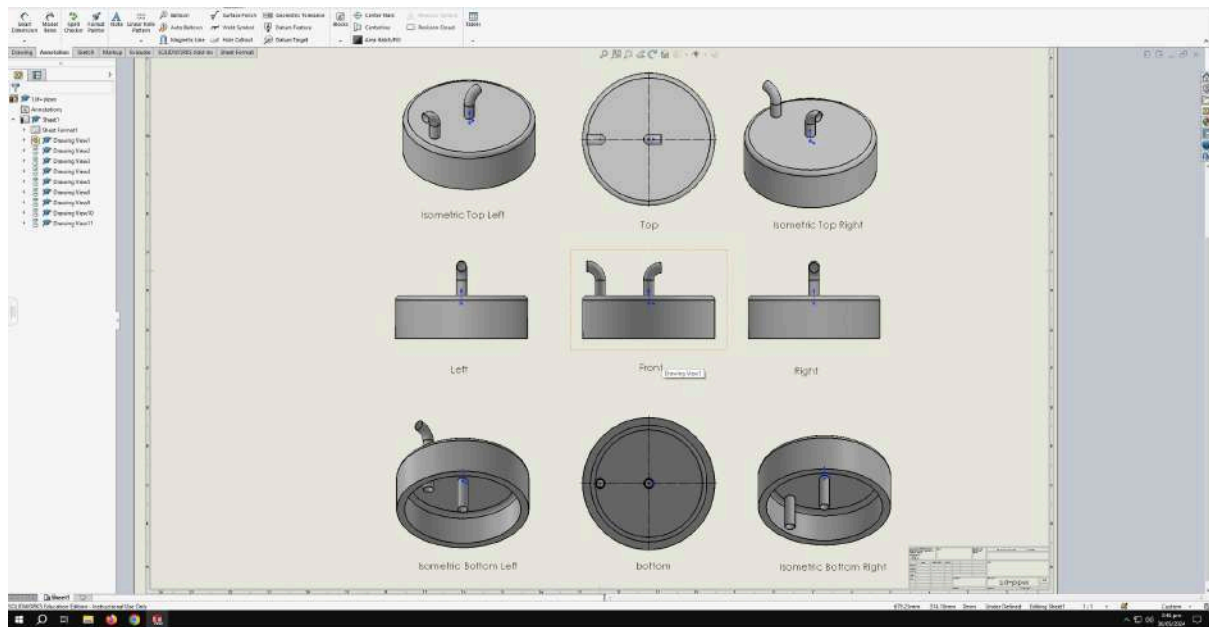
5) I created four new sketches of for how the pipes will curve around so they can then be attached



- 7) Assigning the materials—of the 2 options I have, PVC and steel—after comparing them, I have gone with PVC. this is due to minimal differences in performance, cost and weight (difference of 3 kg, 3532.95 g, compared to 588.83 g)

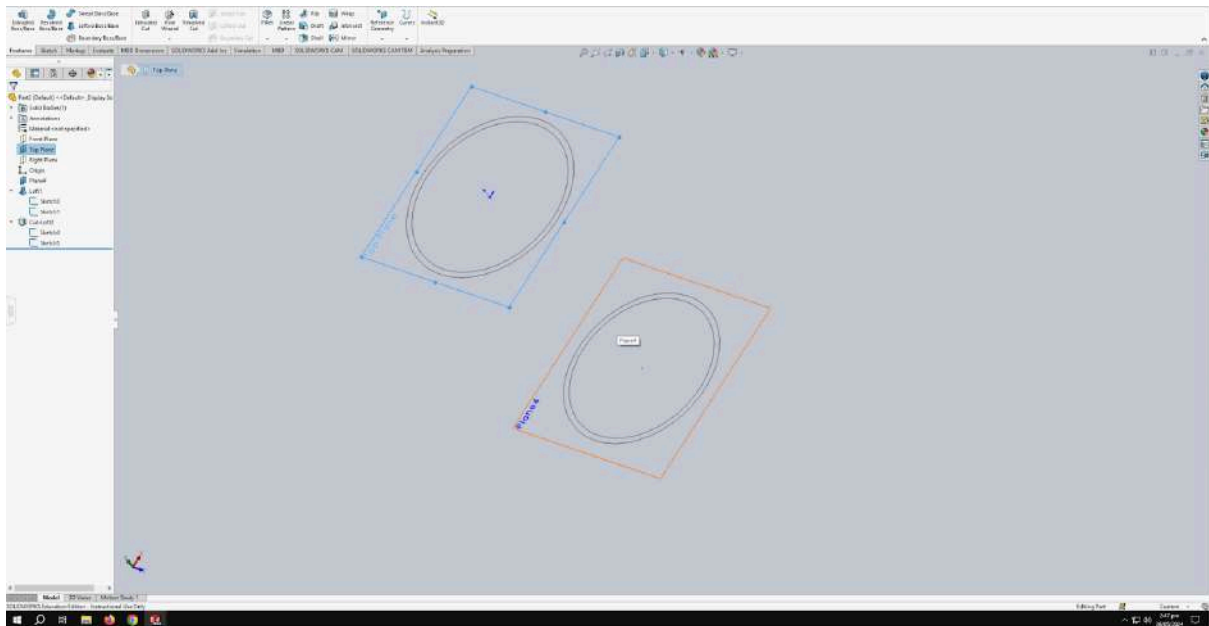


8) Section views: I have made a drawing containing all the necessary Views (left, Top, Right, and Bottom)

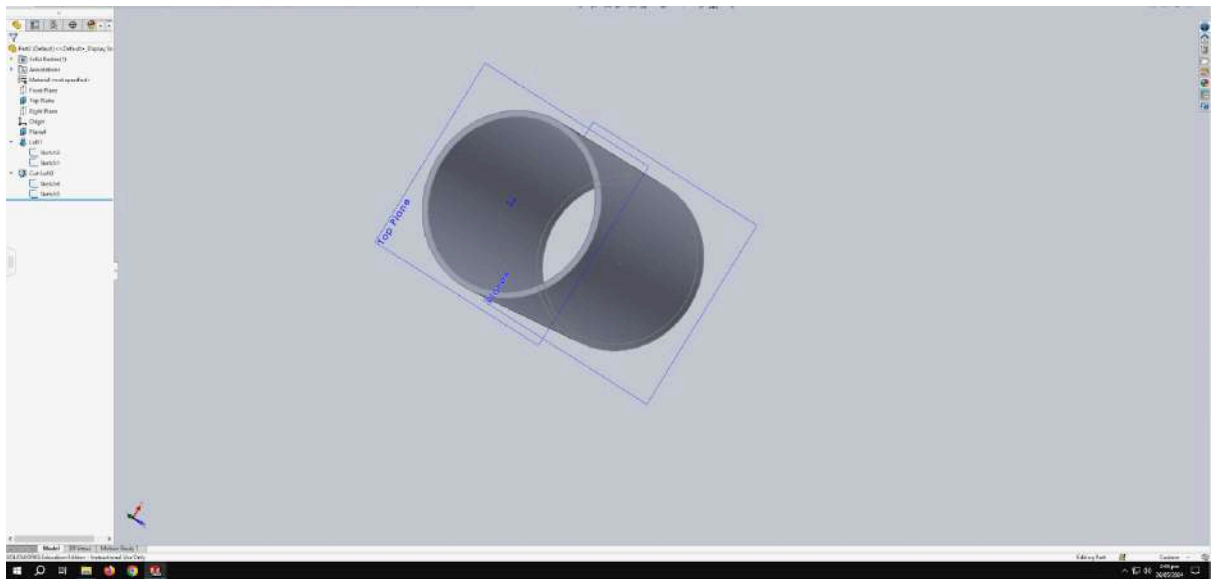


Part 3 - Filter case

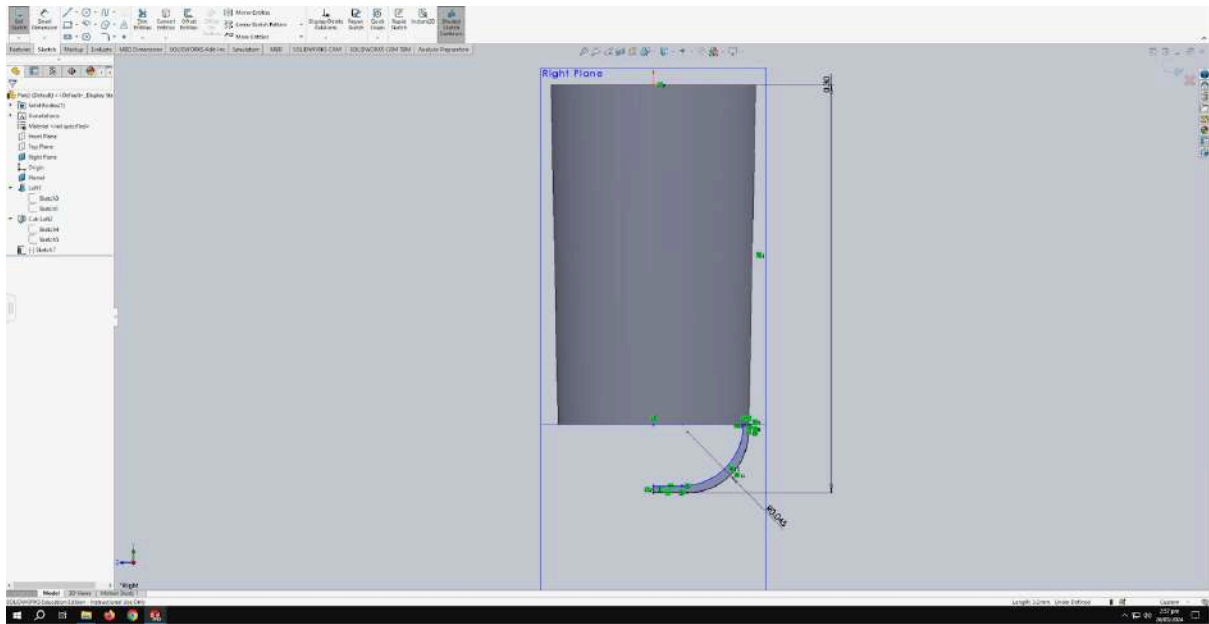
1) Create sketches of case



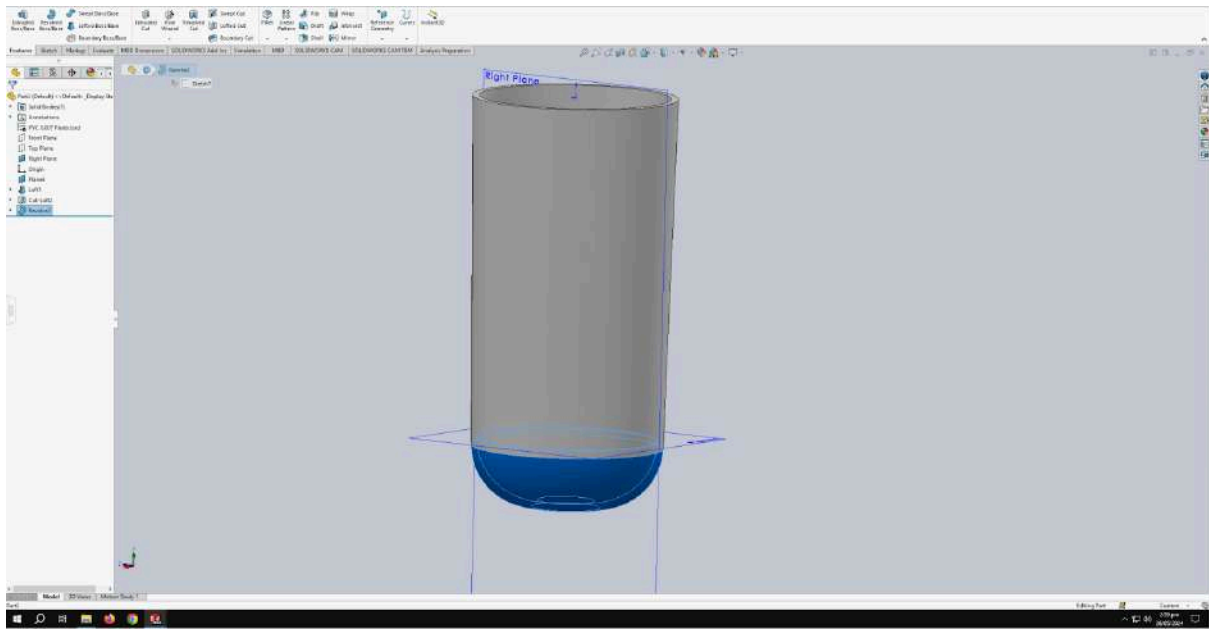
2) Loft boss and cut



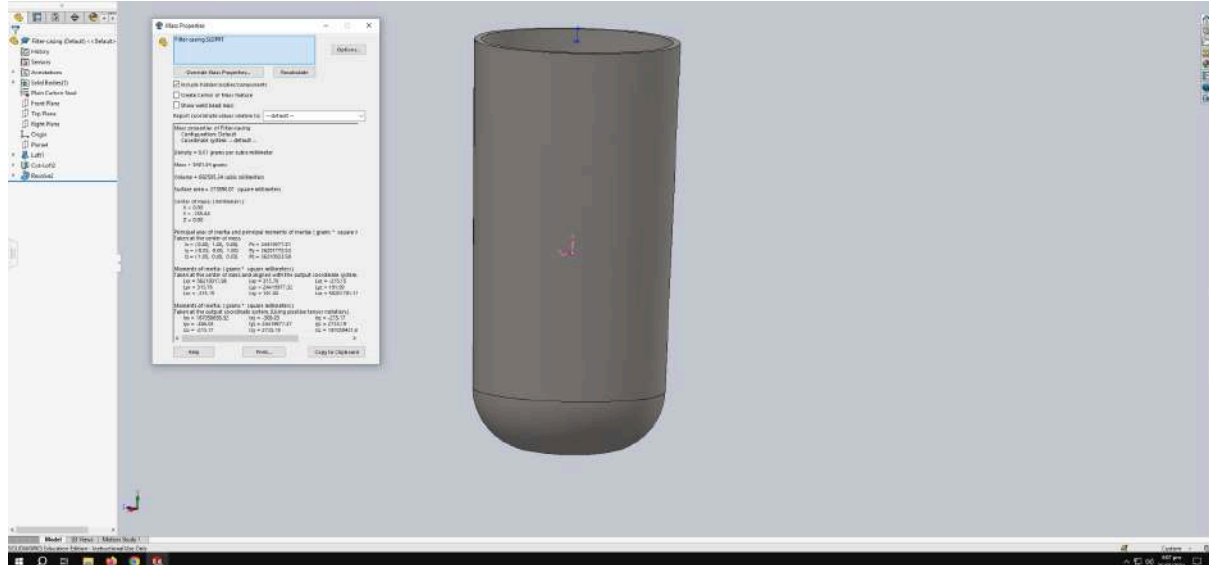
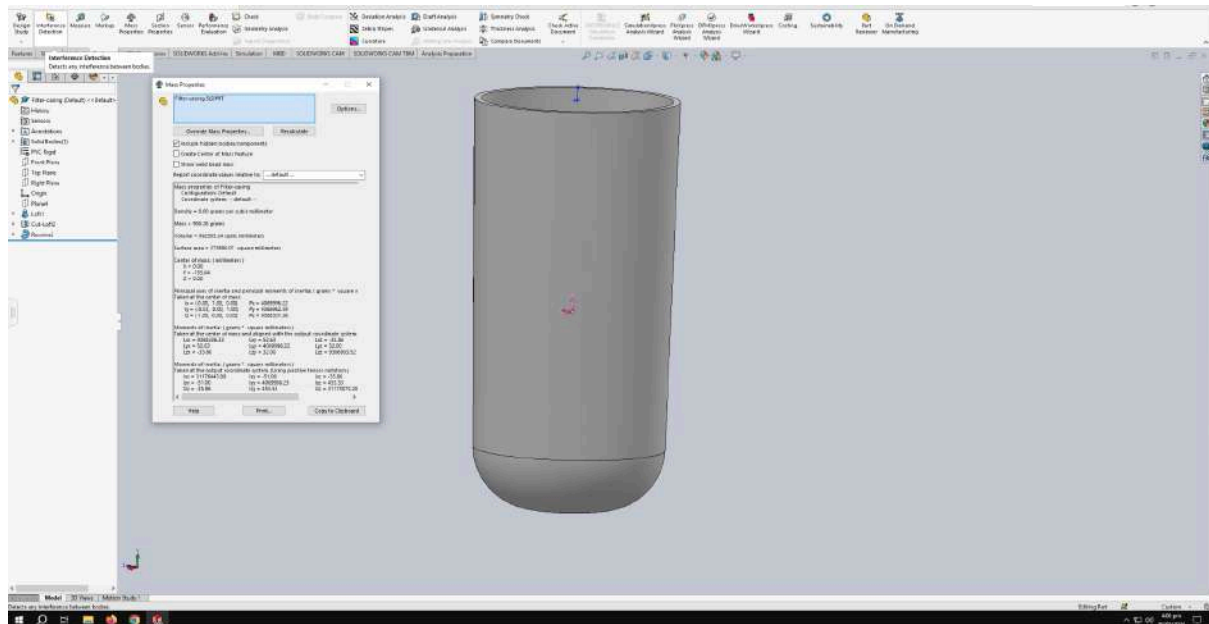
3) Create bottom sketch



4) Revolve bottom around the central axis

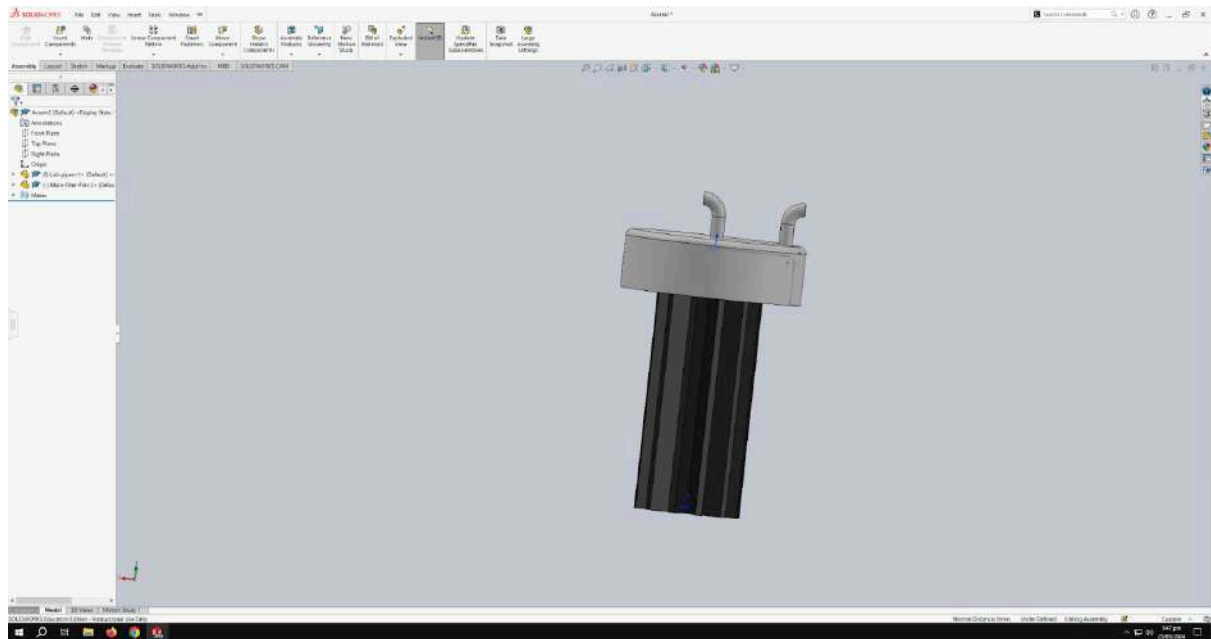


- 5) Compare and choose a material. Like with the lid and pipes, I had the option between pvc and steel; again, I have gone with pvc as it is lighter (900.2 g compared to 5401.5 g) and more cost effective.

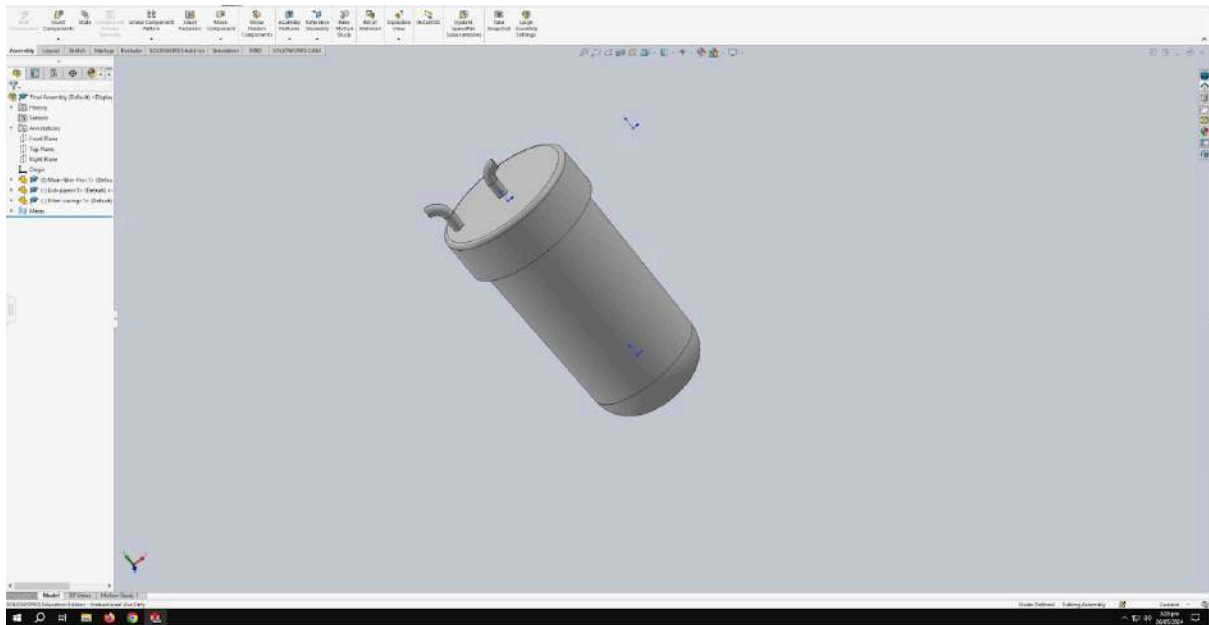


Final Assembly - Pre-filter

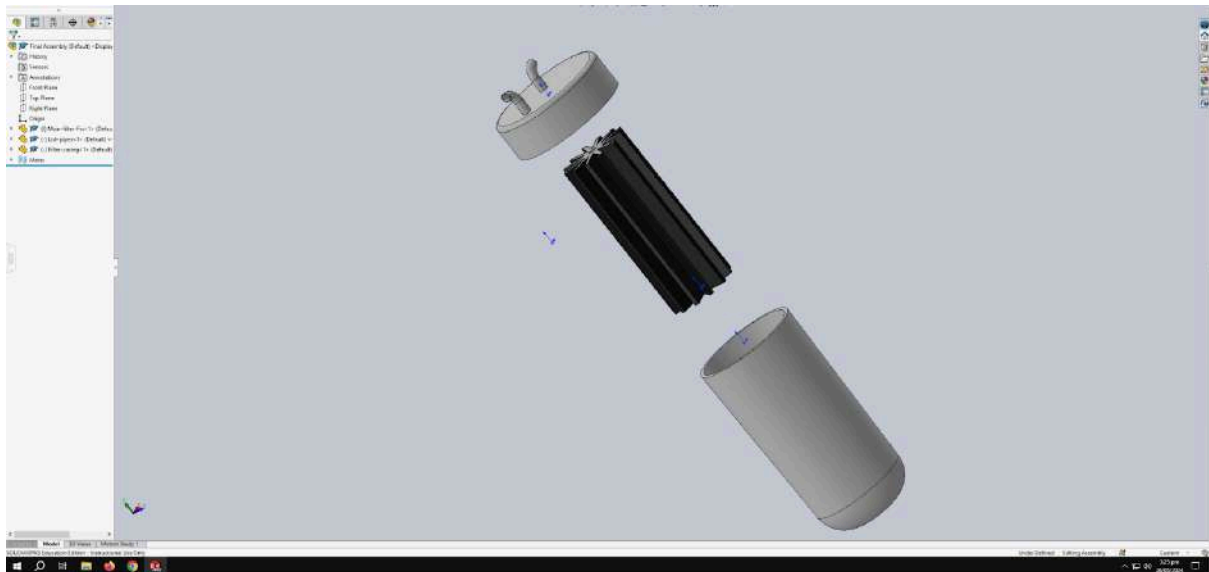
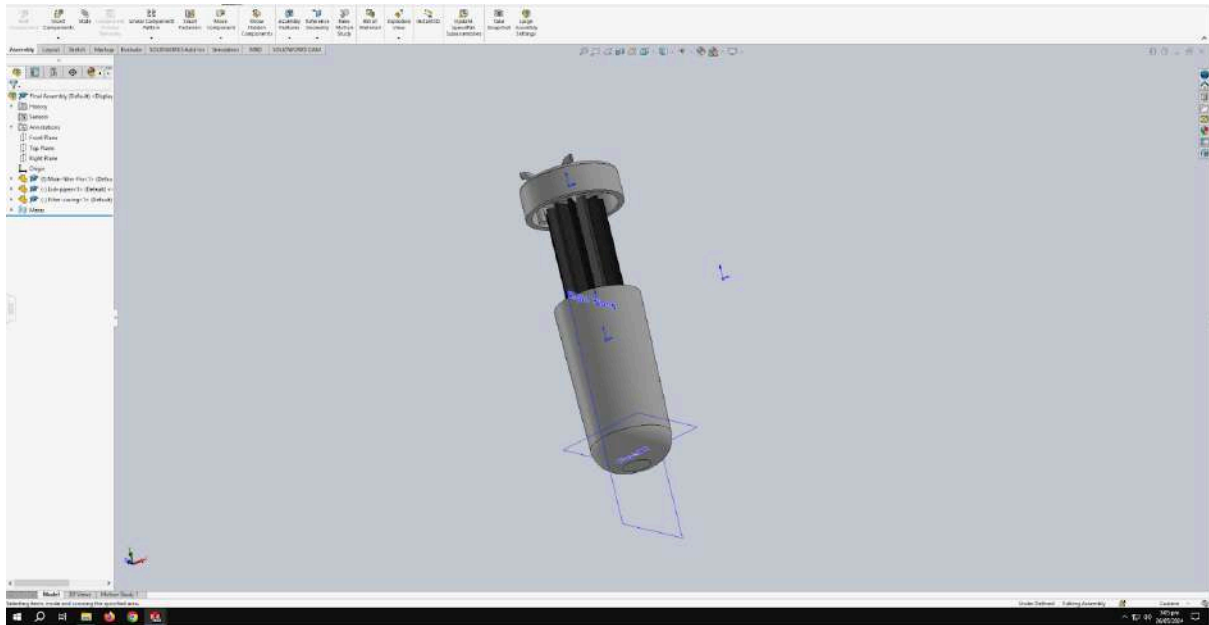
1) Dragged in the lid and filter itself. I mated them together using 2 mates.



2) Dragged in the filter casing. I mated this to the lid and to the front plane to lock rotation, and thus the filter became fully defined.



3) Created 2 exploded views to better understand the workings of the filter.



- 4) I then created a drawing with all the parts and the final assembly so that it is easier to get a better understanding of the pre-filter. Included in this drawing are a bill of materials table, section views of the pre-filter and internal filter, an exploded view of the final pre-filter, and model, projected, and isometric views of all the parts.

